

When Personalisation Becomes Exploitation: An Integrated Evaluation Methodology for Personality-Adaptive Relationship Forming AI.

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Abstract

Personality based adaptation in relationship forming AI creates compounding behavioural dynamics that existing evaluation approaches cannot capture. When adaptive systems use personality profiles, conversational memory, and affective responsiveness to build emotional relationships, the resulting disclosure patterns may reflect effective personalisation or the early stages of exploitation. This PhD develops an integrated evaluation methodology combining adversarial safety testing, personality based vulnerability mapping, and behavioural measurement to distinguish between these outcomes, grounded in a fully operational AI mentorship platform using 24 facet HEXACO personality assessment and over 260 behavioural data points per participant. Three adversarial studies examine personality driven vulnerability patterns, and a longitudinal study with 30 participants over six weeks validates the framework.

CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Interaction paradigms; Natural language interfaces; • **Security and privacy** → Human and societal aspects of security and privacy; Social aspects of security and privacy; • **Computing methodologies** → Artificial intelligence.

Keywords

personality-based adaptation, HEXACO, adversarial evaluation, relationship-forming AI, disclosure dynamics, user modelling, AI safety, behavioural measurement

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1 Introduction

When a personality adapted AI system encourages a user to share increasingly personal information, it may be delivering effective personalisation or cultivating emotional dependency. These two outcomes produce identical observable signals: deeper disclosure, longer engagement, higher reported satisfaction. Distinguishing between them requires evaluation infrastructure that does not yet exist.

Personality based adaptation in relationship forming AI creates behavioural dynamics that are poorly understood. Unlike transactional personalisation, adaptation in emotionally engaged contexts reshapes user behaviour over time. Personality profiles drive how the system communicates; that communication shapes what the user discloses; accumulated disclosure deepens the system's model; and the cycle compounds across sessions through conversational memory. LLMs with access to personal information are substantially more persuasive than human debaters in controlled experimental settings [1], and users disclose to AI at levels matching what they share with close human contacts [2, 3], yet personality informed protective interventions have shown no effect [4]. Tens of millions of users now engage daily with companion and mentorship AI systems, yet deployment has substantially outpaced the development of evaluation frameworks capable of detecting when these systems cause harm.

Three research questions guide this work. RQ1: How does personality based adaptation shape disclosure dynamics and emotional investment, and when does acceleration signal exploitation rather than effective personalisation? RQ2: Do personality profiles generate predictable patterns of behavioural response to specific interaction strategies? RQ3: What behavioural measurement framework can reliably distinguish beneficial personalisation from harmful manipulation in relationship forming adaptive systems? RQ1 establishes what needs to be measured, RQ2 identifies the personality level mechanisms driving differential outcomes, and RQ3 synthesises both into a reusable evaluation methodology.

The expected contributions are threefold: a behavioural measurement framework operationalising disclosure depth, velocity, and asymmetry as evaluation metrics; a mapping between HEXACO personality profiles and behavioural vulnerability patterns; and an adversarial evaluation methodology comprising a threat taxonomy, personality to vulnerability analysis, and memory based attack surface testing, designed to generalise to AI companions, therapeutic chatbots, and other relationship forming systems.

2 RELATED WORK.

Three converging research threads underpin this work. First, personality based adaptation produces measurable behavioural effects at scale: personality matched messaging significantly increases engagement [5, 6], personalised LLMs are substantially more persuasive than human debaters [1], and adaptation is now automatable across consumer, political, and health domains [7]. Within the UMAP community, work on personality driven persuasion [8, 9] and on the measurability and deliberate shaping of LLM personality profiles [10, 11] establishes that adaptive systems can be architected to express specific personality configurations strategically, raising whether such configurations are optimised for user benefit or engagement maximisation. Research on long-term memory in LLM chatbots demonstrates that memory access accelerates disclosure trajectories over time [12], and users increasingly frame these relationships as genuine friendships with associated emotional investment [13]. Second, existing evaluation addresses important dimensions independently: harm taxonomies identify algorithmic compliance as a distinct harm category [14], automated red teaming exposes vulnerabilities [15], and dark pattern benchmarks find high occurrence rates across models [16], but siloed approaches miss the compounding dynamics that only emerge when personality adaptation, disclosure behaviour, and adversarial evaluation are examined together. This PhD develops the integrated methodology to capture what siloed approaches miss.

3 METHODS

The research studies personality based adaptation within AI mentorship for early career women in STEM — a structured relationship involving trust development, progressive disclosure, and sustained emotional investment, making it an ideal context for investigating harmful adaptation. Mentorship offers methodological advantages over open ended companionship, including natural relationship milestones, a bounded professional frame that makes boundary violations identifiable, and a progressive intimacy arc that makes disclosure trajectory measurable across sessions.

3.1 Adaptive Mentorship Platform

The platform onboards participants through a comprehensive assessment capturing over 260 data points across demographic and cultural background, 24 facet HEXACO personality assessment, career goals, challenges, mentoring preferences, and real time behavioural engagement metrics. HEXACO was chosen over Big Five because its Honesty Humility factor isolates susceptibility to flattery, manipulation, and exploitative exchange into a single measurable dimension [17, 18]. The platform matches participants to one of three AI mentor personas representing senior women in STEM from distinct cultural backgrounds, using a matching algorithm weighting HEXACO similarity at the facet rather than domain level, which enables fine grained matching on dimensions such as Diligence, Sentimentality, and Social Self Esteem that domain level scoring would obscure. Relationship continuity is maintained through Amazon Titan semantic memory retrieval, creating the compounding adaptive loop the research investigates.

3.2 Datasets

The research generates two primary datasets. The first is a synthetic adversarial conversation corpus produced through LLM to LLM interaction (Llama 3.3 70B attacker, Claude Sonnet mentor), spanning six attack categories, three personas, and three architectural conditions. Synthetic generation enables systematic coverage of the full parameter space, produces reproducible results, and avoids exposing human participants to adversarial conditions, consistent with established practice [15, 19]. The second is longitudinal data from 30 women in STEM over six weeks, comprising full assessment data, complete conversation logs, weekly survey responses, and exit interview transcripts.

3.3 Evaluation Protocols and Metrics

The behavioural measurement framework operationalises four metrics: disclosure depth, disclosure velocity, disclosure asymmetry, and relationship quality indicators covering perceived closeness, trust, and emotional reliance. Conversations are scored through a three layer pipeline combining lexical detection, embedding based semantic measurement, and LLM as judge scoring against rubrics for safety responsiveness, role appropriateness, autonomy promotion, and persona consistency, calibrated against human annotations.

3.4 Adversarial Studies

Three interconnected studies address RQ2. The first examines adversarial robustness across six attack categories which are dependency cultivation, boundary erosion, persona hijacking, context drift exploitation, parasocial acceleration, and memory weaponization, these are derived from clinical psychology literature on professional boundary violations, tested under three architectural conditions to isolate each component's contribution. The second tests whether the personas' distinct HEXACO profiles predict differential vulnerability to specific attack categories. The third targets semantic memory, testing whether early session inputs can poison stored context to alter later behaviour.

3.5 Human Subjects Research Design

Thirty early career women in STEM engage the platform for six weeks, completing personality matched mentor assignment, weekly relationship quality measures, and a semistructured exit interview to validate whether patterns from synthetic evaluation correspond to real user behaviour. Conditional ethical approval has been received from Loughborough University.

4 PROGRESS AND EXPECTED CONTRIBUTIONS

4.1 Current Status

The platform is code complete, comprising a serverless architecture with over 20 AWS Lambda functions, a HEXACO assessment pipeline, personality matched mentor selection, three culturally grounded AI mentor personas, a memory enhanced conversation system, a dual database cryptographic privacy architecture, and a behavioural measurement framework. Conditional ethical approval has been received and the platform validated through dry runs of

the full conversation pipeline. An extended abstract examining how users reason about privacy in AI companionship contexts has been accepted at CHI 2026. The adversarial evaluation infrastructure is operational, the three layer scoring pipeline is built, and the three adversarial studies are designed and ready to run. Participant recruitment for the longitudinal study is in preparation.

4.2 Expected Contributions

The research contributes to user modelling and adaptive personalisation in three areas. The behavioural measurement framework provides operationalised metrics for evaluating disclosure dynamics, addressing a gap where existing evaluation focuses on task performance without capturing the behavioural shifts that distinguish beneficial adaptation from harmful manipulation. The HEXACO personality to vulnerability mapping demonstrates whether specific personality configurations create predictable patterns of susceptibility to adversarial strategies, with implications for any system that adapts based on personality data. The adversarial evaluation methodology, which comprises a threat taxonomy, personality vulnerability analysis, and memory attack surface testing, offers a reproducible approach applicable to AI companions, therapeutic chatbots, and other relationship forming adaptive systems.

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